Deliverable 1
Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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August 13, 2026

This investigation formalizes the phenomenon of *structural aliasing* within patch-based time-series foundation models, focusing specifically on Amazon's Chronos-Bolt framework.[1, 2] The project focuses on smart-grid load balancing.

Domain Description

To ground this abstract architectural evaluation, the study utilizes a specific physical domain as an empirical reference layer. This reference setting models the solar energy produced by a photovoltaic array attached to an internal load, a local storage battery, and the national electricity grid. The explicit operational objective within this system is to evaluate production variations over short-term future horizons so that the system can dynamically modify the internal load, attach and detach the battery unit, with the aim of maintaining internal grid stability and enabling precise anomaly detection.

Within this infrastructure, a control AI agent is responsible for executing real-time tasks: dynamic load-balancing, battery dispatch, state-monitoring and active anomaly detection, and internal grid stability maintenance.

This analysis is directly intended for researchers, engineers and practitioners utilizing foundation models within the Chronos family for time-series analysis, representation learning, and forecasting, as well as individuals working in the field of energy management who are interested in characterizing the fundamental limitations, structural boundaries, and latent capabilities of these architectures.

Purpose and Scope

This project formalizes and empirically validates the phenomenon of *structural aliasing* in patch-based time-series foundation models, using Amazon Chronos-Bolt-Tiny as the target architecture.

Chronos-Bolt-Tiny operates with a fixed patch size of $P = 16$, stride of $S = 16$ and context window of $T = 2048$ samples as default settings; its architecture comprises four encoder blocks followed by four decoder blocks terminating in a direct quantile projection head.[3, 4]

This study aims to provide a comprehensive understanding of the relationships between raw signal characteristics and the latent representations learned by the model, systematically identifying the critical

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

links between sampling frequency (f_s), patch size (P), and stride (S) in order to propose actionable solutions to mitigate the phenomenon.

The core objective is to challenge the baseline assumption that pre-trained temporal foundation models inherently preserve frequency attributes when processing Nyquist-compliant signals.

Mathematical Formulation

The project provides a formal domain description through OWL2 ontology representation of: the physical system (photovoltaic array, battery storage, national grid, internal load), the signal (frequency, amplitude, timespan, phase) and its patched representation (patch size, stride, sampling frequency). See Appendix A for a formal description of the ontology.

The investigation focuses on the formal definition of the phenomenon of structural aliasing, starting from the tokenization operator and describing phase-locking conditions. The project then formalizes the relationship between the signal frequency, patch size, stride, and sampling frequency, providing a mathematical framework to analyze the conditions under which structural aliasing occurs.

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$. The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{1}$$

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{2}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{3}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π–periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{4}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{5}$$

The same derivation can be applied taking into account translational shifts equal to the patch size P, yielding a similar phase-locking condition

$$2\pi \frac{fP}{f_s} = 2\pi k \Rightarrow x[n+P] = x[n], \qquad k \in \mathbb{N}^+. \tag{6}$$

Solving Equation 4 and Equation 6 for f yields the phase-locking frequency class $\mathcal{F}_{lock}$, which is defined as the set of frequencies producing self-repetition over one stride or over one patch.

$$\mathcal{F}_{lock} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S} \vee f = k\frac{f_s}{P},\ c, k \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{7}$$

The derivation above shows that some translations leave a sinusoid unchanged, this repetition is not evidence that two distinct signals become indistinguishable to the model.

Structural aliasing is therefore treated as an emergent, falsifiable property of the trained representations[5]. At a phase-lock consecutive patch tokens coincide, $\mathbf{t}_k = \mathbf{t}_{k+1}$, so distinct inputs can share a token sequence. We diagnose the phenomenon by *readability* rather than by a fixed distance between representations: structural aliasing is present when the signal's frequency can no longer be recovered from the model's internal state at a locked frequency as well as it can at neighbouring, non-locked ones.

Aliasing implications

The above derivation reframes $\mathcal{F}_{lock}$ as the candidate class where such collisions are hypothesised to occur. In particular, the project tests the following hypotheses.

H1: if $f \in [f_k - \varepsilon, f_k + \varepsilon]$, where $f_k \in \mathcal{F}_{lock}$, the model recovers less information. We expect: higher MDL codelength and lower forecast amplitude recovery than at fixed controls $f_k \pm \delta$, $\delta \gg \varepsilon$. Absent or non-localized loss refutes H1.

H2: the magnitude of degradation at a locked frequency $f_k \in \mathcal{F}_{lock}$ is a property of the stride/patch alignment, not of the signal's own phase: across the S_{f_k} phase offsets already sampled at f_k, the accuracy drop relative to the $f_k \pm \delta$ control is expected to stay approximately constant. A drop whose size varies systematically with phase, rather than only with frequency, refutes H2.

H3: at fixed P, degradation sites at $f = cf_s/S \in \mathcal{F}_{lock}$ move proportionally to $1/S$; at fixed S, sites at $f = kf_s/P \in \mathcal{F}_{lock}$ move proportionally to $1/P$. A site that does not move when the parameter generating it is varied refutes H3.

Data and Evaluation

Two sets of synthetic signals are used to evaluate the presence of structural aliasing in the new models. These include:

- **Sine Wave Signals:** simple sine waves with known frequency systematically varied between 2 Hz and 250 Hz with 1 Hz increments, to assess the model's ability to capture and represent periodic components across a range of frequencies.
- **Time Series Mixup Signals:** 100 signals generated by a sinusoidal mixer based on the TSMixup procedure[1], mixing multiple sine waves of different frequencies; to evaluate the model's capacity to disentangle and represent complex frequency interactions.

Optionally, if previous analyses show evidence of aliasing, the project may analyze test signals from a third set, including 100 signals generated using the KernelSynth[1] tool, to test whether the phenomenon persists in more complex signals. The project then focuses on the analysis of real-world data, through the use of the `Dataset-SolarTechLab.csv` dataset, which contains a one-minute-resolution multivariate telemetry stream of a photovoltaic array[6].

All tests are undertaken for the entire set of re-trained models, to evaluate the impact of stride and patch size on the model's ability to capture and represent frequency information in more complex signals.

Run	Patch (P)	Stride (S)	overlap	#patch @2048	steps	time
1	16	8	0.50	255	100k	2.6 h
2	16	12	0.25	170	100k	2.8 h
3	16	16	0.00	128	100k	pending
4	8	8	0.00	256	100k	2.7 h
5	24	24	0.00	86	100k	2.8 h

Table 1: *Patch/stride configurations retrained from scratch (seed 42) under a uniform* 100*k-step budget;* time *is the wall-clock on the RTX 5060 Laptop GPU. The P*=16*, S*=16 *baseline is retrained under the* same *budget as the variants, so the stride comparison is not confounded by training budget. The official Amazon checkpoint (*200*k steps, full corpus) is used separately, only as an external forecast-quality reference (Figure 1).*

During the analysis a seed is set to ensure reproducibility of the results. All signals considered are sampled at $f_s = 512\,\mathrm{Hz}$.

Methodology

The default `chronos-bolt-tiny` checkpoint, at P=16, S=16, is evaluated first on the sine-wave and mixup sets described above.

Since any change to P or S requires retraining, we retrain a restricted grid around the default $P = 16$, $S = 16$ baseline (see Table 1): two reduced strides at fixed $P = 16$ isolate the effect of patch overlap, while two contiguous patch sizes isolate that of patch width. We exclude $S = 4$: its stride-lock class $\{128, 256, \dots\}$ has one member inside the valid forecast band ($<$ 256Hz according to Nyquist), and therefore yields no informative local $H1/H3$ contrast on the integer-Hz grid.

Configurations with $S > P$ are excluded by construction: consecutive patches would be separated by $S - P$ unobserved samples, so any loss of frequency information could be attributed to discarded data rather than to tokenization geometry. The requested symmetry is therefore realised across the zero-overlap configurations, with $P = S \in \{8, 16, 24\}$ bracketing the default.

Parameter	**Adopted**	**Chronos official**	**Max / constraint**
Architecture (fixed, from Bolt-tiny)			
`context_length`	2048	2048	Bolt-tiny
`prediction_length`	64	64	Bolt-tiny
quantile levels	9	9	Bolt-tiny
Optimisation			
`batch_size`	32	32	VRAM 8 GB
`max_steps`	100 000	200 000	compute budget
`learning_rate`	10^{-3}	10^{-3}	—
scheduler / warmup	linear / 0	linear / 0	—
`weight_decay`	0.0	0.0	—
`grad_clip_norm`	1.0	1.0	—
optimiser / precision	AdamW (fused) / fp32+TF32	—	
Data pipeline			
corpora (streaming)	TSMixup 10M + KernelSynth 1M		official
mixing ratio	9:1	9:1	—
`shuffle_buffer`	10 000	100 000	streaming timeout
`min_past` / `max_missing` / `drop_prob`	60 / 0.9 / 0.2	—	

Table 2: *Training hyperparameters: adopted value, official Chronos recipe (`chronos-t5-tiny`), and usable maximum / constraint. Only (P, S), the step budget (*100*k vs.* 200*k) and the shuffle buffer (*10*k vs.* 100*k) depart from the official values, to fit the single-GPU budget.*

Each (P, S) configuration is retrained from scratch: changing P resizes the first patch-embedding block and changing S affects the learned inter-patch dynamics, so weights are not transferable. Loading only the `chronos-bolt-tiny` architecture, the loop follows the *published Chronos-T5* training recipe (fused AdamW, linear LR 10^{-3}, `fp32+TF32`, masked quantile loss) on the two official corpora, TSMixup (10M) and KernelSynth (1M) streamed at the 9:1 ratio[7]; the complete Chronos-Bolt recipe is not publicly released, so this is a budget-matched adaptation of the published hyperparameters rather than an exact reproduction. Every hyperparameter stays at its published value (see Table 2); only (P, S) varies, and a uniform 100k-step budget (vs. 200k) is applied to *all* configurations, including the $P{=}S{=}16$ baseline, so the comparison is not confounded by training budget. A single seed (42) is used for now; multiple seeds remain a planned extension.

Training ran on one Alienware 16 Aurora (NVIDIA RTX 5060 Laptop GPU, 8 GB VRAM; 32 GB RAM), ≈11 h in total. Forecasting quality of the retrained models is shown in Figure 1. If necessary, the project may extend the training budget to the full parameters.

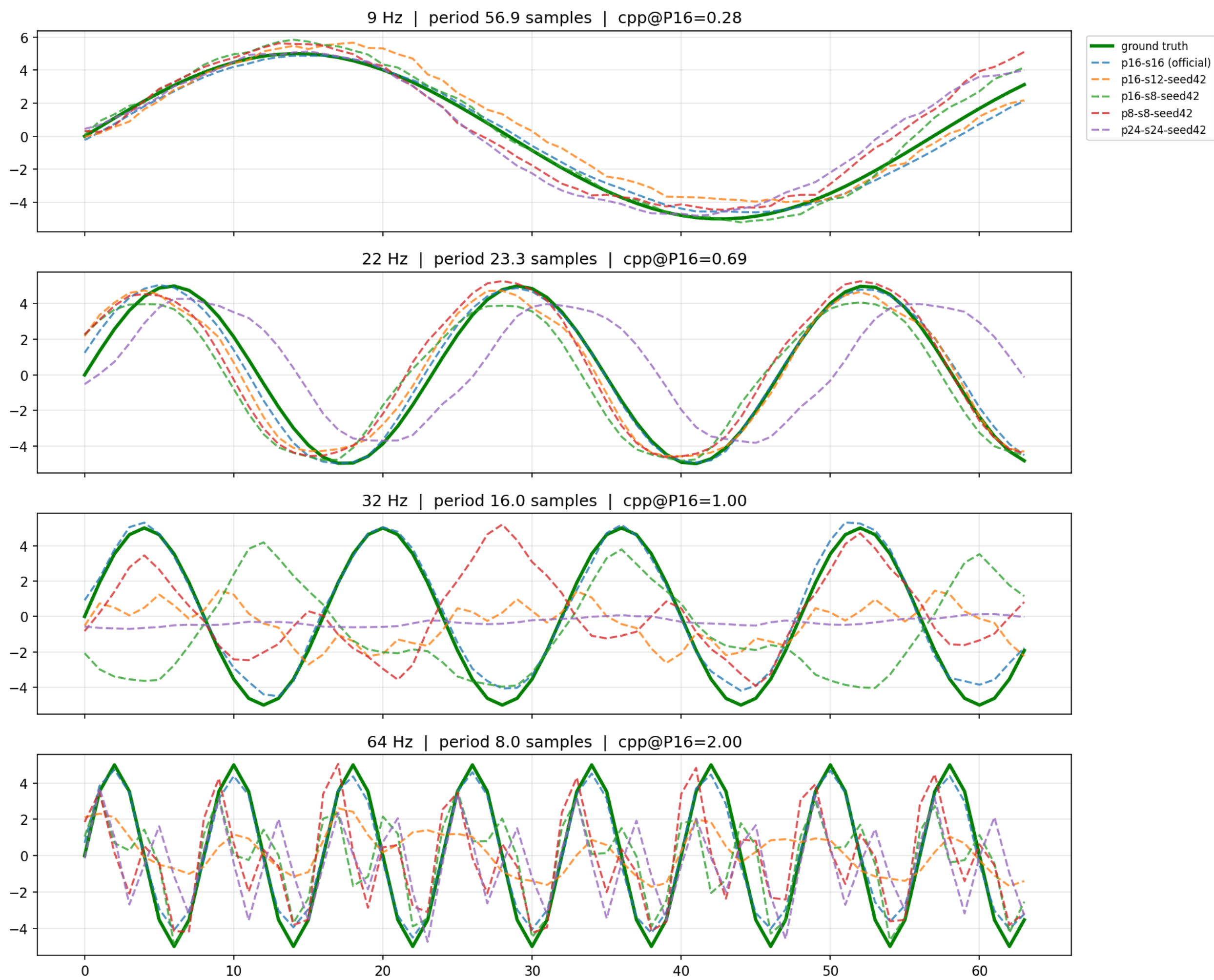

Figure 1: *Forecast quality of the retrained models. Per frequency (rows), each 64-step forecast (dashed) vs. ground truth (green); titles give cycles-per-patch at $P = 16$ (cpp). The real model performances are better than the retrained, due to the larger training budget and more extensive corpora.*

Probing Methodology

Following MDL probing[8], the 256-dimensional [REG] representation is captured after each encoder block and each decoder block, plus two points at the output stage: the pre-projection 256-d [REG] input, and the projected quantile output (Pagani et al.'s h_4[5]), where the compression drop reported in the source paper is concentrated. Probing this single per-block vector, rather than the full flattened patch sequence, avoids a dimensionality mismatch across stages and patch/stride variants.

All probe points are compared on the same footing via a prequential-MDL codelength $L(D)$, reported as compression $\mathrm{SV} = 1 - L(D)/L_{\mathrm{uniform}}(D)$, computed independently per probe point for each of the seven hierarchical band-classification tasks[5]. Each value is benchmarked against a shuffled-label control ($\mathrm{SV} \le 0$ confirms the probe is not memorising noise) and a random-initialisation control (identical untrained architecture, isolating learned information from architectural artefacts).

Independently, a dense per-frequency classification-accuracy profile is obtained at the projected output by sweeping the same band-classification probe across the operational spectrum, with non-redundant phase sampling $S_f = \min\{f_s/\gcd(f, f_s) - 1,\, N\}$ replacing independently drawn random phases.

Bayesian Analysis

Each hypothesis receives its own estimand and model: two for the *size* of the phase-lock effect, behavioural and representational ($H1$), one for its dependence on phase ($H2$), one for its *location* in frequency ($H3$). All are restricted to the range over which valid forecasts are observed, expected to be a subset of $[2, 250]\,\mathrm{Hz}$.

Local Contrast Analysis (H1, behavioural): to test localized recovery drop at stride-lock, signals at exact $f_k \in \mathcal{F}_{lock}$ are paired with controls at $f_k \pm 0.25 f_s/S$ sharing phase, amplitude, noise level and background. Letting R denote the forecast amplitude recovery ratio, the paired logarithmic local contrast is:

$$d = \log(R_k + 0.01) - \frac{\log(R_- + 0.01) + \log(R_+ + 0.01)}{2} \tag{8}$$

with $d < 0$ indicating localized attenuation at the lock. Contrasts are modeled using a heavy-tailed Student-$t_4(\mu_i, \sigma^2)$ likelihood over a three-level hierarchy,

$$\mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \qquad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c,\ \tau\Big), \qquad O_c = \frac{P_c - S_c}{P_c}, \tag{9}$$

where c indexes configurations, k the lock harmonic, b the background realisation, and $\widetilde{O}_c$ the centred patch overlap. The population effect $\bar{\beta}$ and the overlap slope δ_O receive Student-$t_4(0, 0.5)$ priors, centered at zero due to the absence of prior quantitative estimates in the literature; on this log-ratio scale 0.5 admits a factor $e^{0.5} \approx 1.65$. All scales receive Half-Student-$t_4(0, 0.5)$ priors, and with one training seed inference is conditional on that checkpoint.

Absolute Performance Analysis (H1, representational): to evaluate a raw non-negative metric, specifically the prequential MDL codelength $L(D)$, we use a Bayesian Gamma regression model with a log link:

$$y_i \sim \mathrm{Gamma}(\mathrm{shape} = k,\, \mathrm{mean} = \mu_i), \qquad \log(\mu_i) = \alpha_0 + \theta_{\mathrm{lock}} \cdot \mathrm{IsLocked} \tag{10}$$

where $\mathrm{IsLocked} \in \{0, 1\}$ flags phase-lock conditions, and $\theta_{\mathrm{lock}} \sim \mathcal{N}(0, 0.5)$ captures the relative codelength expansion. The Gamma shape parameter receives a weakly informative prior, $k \sim \mathrm{Gamma}(2, 0.1)$, allowing the model to adaptively capture metric dispersion.

The seven band tasks of subsection span the whole band, so one $L(D)$ per probe point leaves IsLocked nothing to attach to. We therefore compute $L(D)$ *per frequency*: at each candidate f_c a balanced probe separates tones at $f_c \pm 1\,\mathrm{Hz}$ from the same [REG] vector, under an unchanged prequential protocol. Band-task values remain a descriptive cross-check.

Phase-Invariance Analysis (H2): Equation 8 averages the S_{f_k} phase offsets sampled at f_k and therefore cannot falsify a claim that is specifically about phase. Retaining the phase index, we cut the phase circle into eight equal bins, give each bin its own offset u_p, and model the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$ as

$$\mathbb{E}[y_i] = \beta_{c[i]} + u_{k[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi), \tag{11}$$

under the same likelihood, with $\sigma_\phi \sim \text{Half-}\mathcal{N}(0, 0.25)$. The estimand σ_ϕ is the spread of those offsets: how much the deficit moves as the signal slides through its cycle. $H2$ states the deficit is set by the stride/patch alignment rather than the signal's own phase, i.e. $\sigma_\phi \approx 0$; we report $\Pr(\sigma_\phi < \log 1.1 \mid D)$ and a leave-one-out comparison against the phase-free model. A variance component assumes nothing about the *shape* of the dependence: any systematic variation with phase inflates σ_ϕ.

Site-Geometry Analysis (H3): $H3$ concerns *where* the degradation sits, so it is tested on the token-collapse profile $z_g(f)$, the across-patch dispersion of the input-patch embeddings. All geometries $g = (P, S)$ share one sweep grid, a uniform 1 Hz grid unioned with the predicted sites of *every* configuration, so none is scored on a grid privileging its own prediction. Labelling each frequency by the grid it belongs to,

$$\log z_g(f) \sim \mathcal{N}\left(\alpha_g + \theta_S \mathbb{1}[f \in \{c f_s/S\}] + \theta_P \mathbb{1}[f \in \{k f_s/P\}],\ \sigma\right), \tag{12}$$

with $\alpha_g, \theta_S, \theta_P \sim \mathcal{N}(0, 1)$; $H3$ predicts $\theta_S < 0$. A leave-one-out comparison of each indicator alone against a null identifies which parameter generates the sites, the $S < P$ configurations being what separates the two families. Movement is then estimated as a plain regression of the *measured* comb spacing $\hat{\Delta}_g$ (median successive difference between detected sites) on the two *predicted* spacings,

$$\hat{\Delta}_g \sim \mathcal{N}\left(\kappa_0 + \kappa_S \frac{f_s}{S_g} + \kappa_P \frac{f_s}{P_g},\ \sigma\right), \tag{13}$$

with $\kappa_S, \kappa_P \sim \mathcal{N}(0, 1)$ and $\kappa_0 \sim \mathcal{N}(0, 10)\,\text{Hz}$. $H3$ predicts $\kappa_S = 1$ and $\kappa_P = 0$: the spacing tracks the stride prediction one-for-one and ignores the patch one, assessed against the ROPE $|\kappa_S - 1| < 0.1$. Since $\hat{\Delta}_g$ is read off a discrete sweep, the residual scale enters as $\sqrt{\sigma^2 + \Delta f^2}$; without that floor, sites landing exactly on the predicted comb would drive σ to zero and degenerate the posterior. Because the design varies P only along $P = S$ (Table 1), κ_P stays partially confounded and the fixed-P=16 series carries the identification.

Inference: each model reports its effect on the natural scale with a 95% credible interval, the probability of at least 20% attenuation $\Pr(\bar{\beta} < \log 0.8 \mid D)$, and the practical-equivalence probability $\Pr(|\bar{\beta}| < \log 1.1 \mid D)$; competing formulations are compared by leave-one-out cross-validation rather than by null-hypothesis testing. Fitting uses NUTS (4 chains, 2000 warm-up and 2000 draws), accepted only at $\hat{R} < 1.01$, effective sample sizes above 1000 and no divergences. The workflow runs prior predictive checks before data collection, a simulation-based parameter-recovery check, posterior predictive checks stratified by frequency, phase, P, S and generator, and prior sensitivity over the scales $\{0.25, 0.5, 1.0\}$.

Security Implications and Risk Assessment

Finally, the project includes a formal risk assessment grounded in the NIST AI Risk Management Framework (AI RMF)[9] to evaluate structural aliasing as a systematic vulnerability within cyber-physical energy systems. This analysis outlines plausible threat models and attack surfaces, possible mitigation strategies, and an evaluation of whether the proposed mitigation strategies effectively reduce residual exposure for the automated control AI agent.

Reducing the stride raises the lowest member f_s/S of the stride branch of $\mathcal{F}_{lock}$ and thins its members below Nyquist; the reduced-stride configurations $S = 12$ and $S = 8$ at fixed $P = 16$ test this mitigation against the S=16 baseline. The `pv:AliasingEvent` class (see Appendix A) lets the control agent act on the residual risk: a forecast frequency in $\mathcal{F}_{lock}$ triggers a reduced-confidence flag.

The risk assessment is structured around the following key points: context and attack surface, threat modeling, impact, mitigation and residual exposure.

Significance and Impact

This subject is relevant because it explores an under-examined systemic vulnerability native to patch-based temporal foundation architectures.[5] By systematically characterizing the conditions under which structural aliasing emerges, this work provides critical insights into the fundamental limitations of these models, directly informing the design of more robust architectures and training paradigms.

For professionals operating within this domain, the analysis could expose how algorithmic data-deletion anomalies can propagate through automated distribution pipelines. By evaluating how hidden token-space representation collapse can deceive a load-balancing AI agent into executing misaligned control choices, this work could provide insights needed for high-reliability, fault-tolerant smart-grid deployments.

The impact of this project could extend beyond theoretical understanding, as it has direct implications for the reliability and safety of real-world applications that depend on accurate time-series forecasting and representation learning, particularly in high-stakes domains such as energy management.

Appendix A: OWL2 Ontology

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (`pv:PhysicalComponent`):** represents the hardware infrastructure of the microgrid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.
- **Cognitive Layer (`pv:ControlAgent`):** the artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (`pv:Signal`, `pv:PatchedSignal`):** tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (`pv:WeatherObservation`):** captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** dynamic metadata classified into generation (`pv:GenerationState`), systemic disruptions (`pv:AnomalyState`), control directives (`pv:AgentAction`), and signal degradation hazards (`pv:AliasingEvent`).

Generation States

A formal description of the state of the photovoltaic array is provided through the following axioms, which define the generation state based on the measured tilted irradiance in $\mathrm{W/m^2}$:

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \land \texttt{measures}(?pv, ?s) \land \texttt{tiltedIrradianceWm2}(?s, ?g) \land (?g \geq 800.0)\\ &\qquad \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{ClearSky})\end{aligned} \tag{14}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \land \texttt{measures}(?pv, ?s) \land{} & \texttt{tiltedIrradianceWm2}(?s, ?g)\\ & \land (?g \geq 150.0) \land (?g < 800.0)\\ & \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{PartialCloud})\end{aligned} \tag{15}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \land \texttt{measures}(?pv, ?s) \land{} & \texttt{tiltedIrradianceWm2}(?s, ?g)\\ & \land (?g > 0.0) \land (?g < 150.0)\\ & \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Overcast})\end{aligned} \tag{16}$$

$$\begin{aligned}&\texttt{PhotovoltaicArray}(?pv) \land \texttt{measures}(?pv, ?s) \land \texttt{tiltedIrradianceWm2}(?s, ?g) \land (?g \leq 0.0)\\ &\qquad \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Nighttime})\end{aligned} \tag{17}$$

Anomaly States

Formal description of the anomaly states which could be detected in the photovoltaic array, based on the measured irradiance and the measured power.

Aliasing Event

Formal description of the aliasing event, detected when the frequency of the signal is part of the frequency class $\mathcal{F}_{\text{lock}}$.

Agent Actions

Formal description of the action taken by the control agent, based on the current state and aliasing event.

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